



A Research Agenda for Mixed Reality in Automated Vehicles

Andreas Riegler
University of Applied Sciences Upper
Austria
Hagenberg, Austria
Johannes Kepler University Linz
Linz, Austria
andreas.riegler@fh-hagenberg.at

Andreas Riener
Technische Hochschule Ingolstadt
Ingolstadt, Germany
andreas.riener@thi.de

Clemens Holzmann
University of Applied Sciences Upper
Austria
Hagenberg, Austria
clemens.holzmann@fh-ooe.at

ABSTRACT

With the increasing knowledge advancements and availability of mixed reality (MR), the number of its purposes and applications in vehicles rises. Mixed reality may help to increase road safety, support more immersive (non-) driving related activities, and finally enhance driving and passenger experience. MR may also be the enabling technology to increase trust and acceptance in automated vehicles and therefore help on the transition towards automated driving. Further, automated driving extends use cases of virtual reality and other immersive technologies. However, there are still a number of challenges with the use of mixed reality when applied in vehicles, and also several human factors issues need to be solved. This paper aims at presenting a research agenda for using mixed reality technology for automotive user interfaces (UIs) by identifying opportunities and challenges.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; **Virtual reality**; **Interaction techniques**; *User studies*; *Scenario-based design*; *Interface design prototyping*.

KEYWORDS

research agenda, mixed reality, automotive user interfaces, automated driving

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1 INTRODUCTION

Since 2014, the number of scientific publications on mixed reality in automated driving has increased rapidly (see Figure 1). We searched for augmented, virtual and mixed reality in automated driving on ScienceDirect, ACM Digital Library and IEEE Xplore

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and found more than 270 papers. Specifically, the AutomotiveUI community has published approx. 100 research papers (including full and short papers, works in progress, workshops and demos) utilizing mixed reality in vehicles, and shows an increasing trend (2014: 15 publications, 2019: 47 publications).

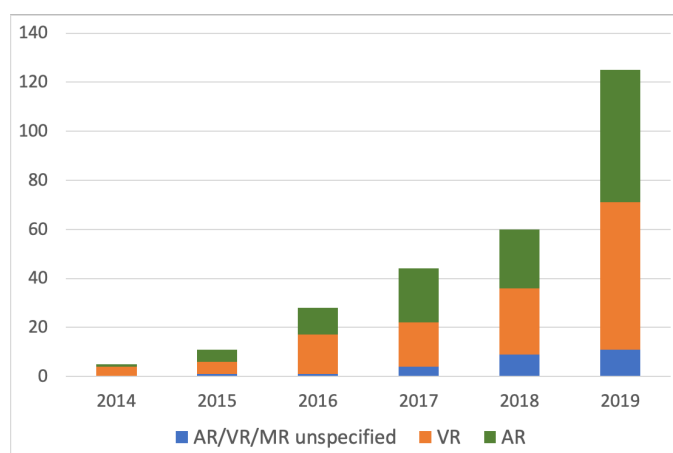
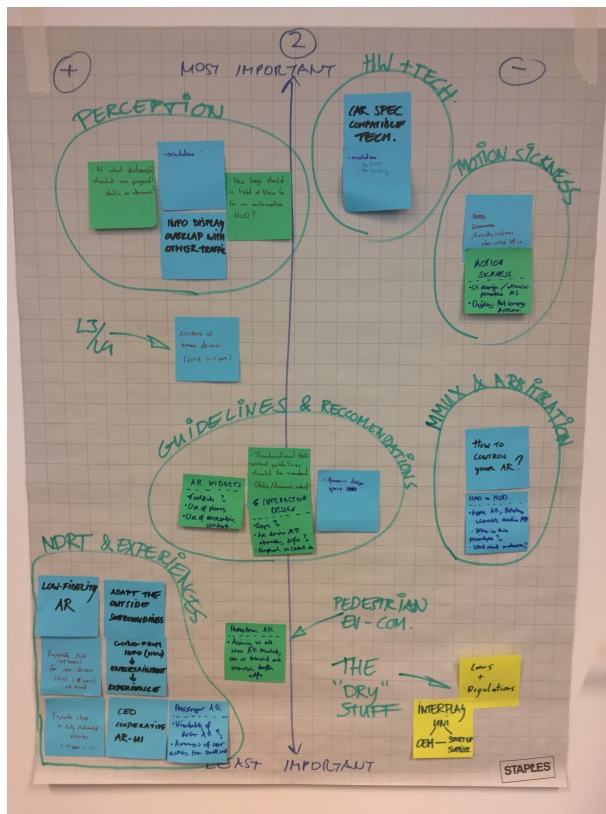
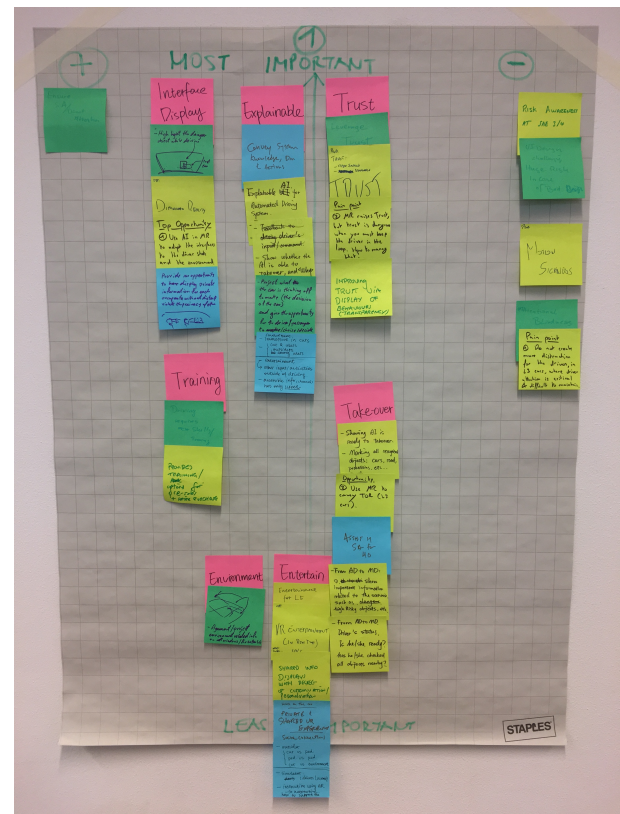


Figure 1: Scientific publications on mixed reality in automated driving, 2014-2019.

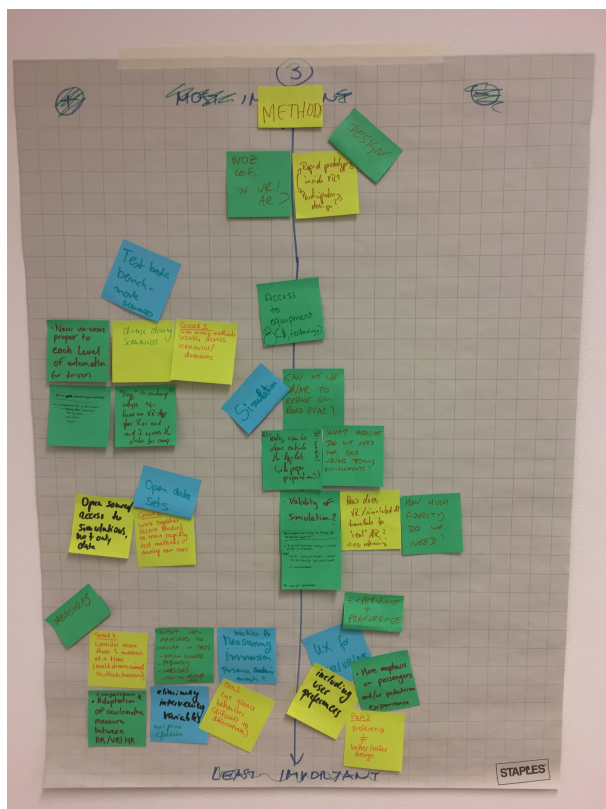
The reality-virtuality continuum [91] describes different levels of reality, from the real world to a fully immersive virtual environment, the so-called virtual reality (VR), which solely consists of 3D digital objects [139]. Mixed reality (MR) encompasses all variations between the real world and the virtual environment. As such, augmented reality (AR) is a technology in which virtual objects are integrated into the real world in real time [6]. The emergence of the Microsoft HoloLens, Magic Leap and similar AR technology allows to research in-vehicle use of AR in a more immersive way [69], thereby enabling researches to explore human-machine interaction concepts more realistically using these AR devices in both lab as well as field studies. Advances in vehicle automation, such as the latest Audi A8 equipped with SAE Level 3 (L3) assistance, provide further use cases for AR applications to support work and entertainment related activities [113, 151] or increase trust in automated driving [49, 153]. Additionally, VR head-mounted displays (HMDs) such as the HTC Vive and Oculus Quest, in conjunction with interaction tracking technologies like Leap Motion for hand tracking and Ultraleap Stratos for haptics provide immersive experiences in



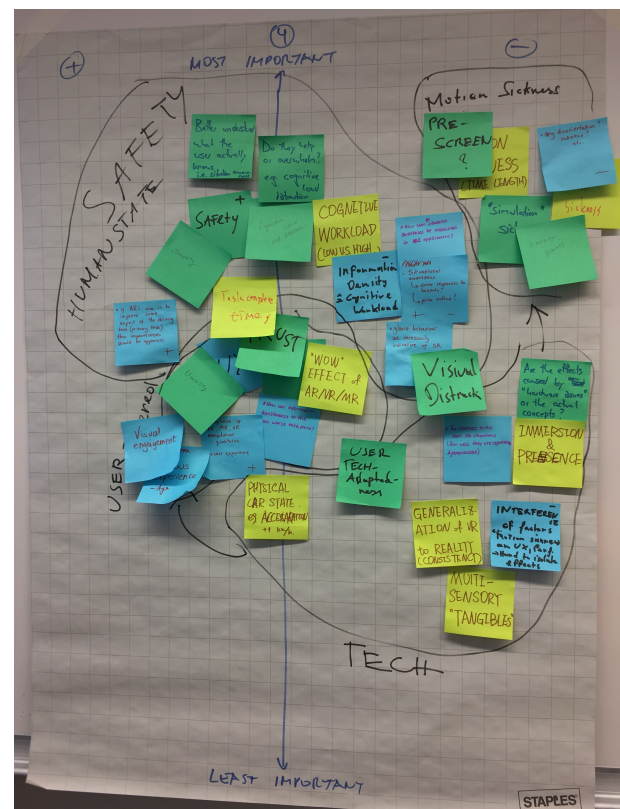
(a) Goals and opportunities.



(b) Accessibility challenges.



(c) Usability testing challenges.



(d) Evaluation challenges.

Figure 2: Results of the workshop on MR in intelligent vehicles.

Table 1: MR applications for SAE level 3 and higher automated driving.

AR	VR
Infotainment/Gamification [20, 76, 125, 138, 140, 148]	Pedestrian interaction [10, 23, 25, 32, 44, 54, 79, 83, 90, 99]
Navigation [7, 12, 12, 121, 141]	Driving simulation [4, 9, 41, 42, 52, 77, 112, 124, 144]
Trust [50, 146, 152, 153]	Feedback/Situation awareness [3, 53, 78, 81, 150]
Feedback/Situation awareness [64, 70, 71, 123]	Tele-operated driving [9, 14, 40, 98]
Mobile office [59, 120, 134, 145]	Passenger experiences [11, 65, 87]
Hazard indication [45, 143]	Navigation [66, 142]
Depth perception [82, 84]	Trust [92, 129, 146]
Task switching [35]	Motion sickness [31, 57]
Passenger experiences [46]	Driving training [34, 137]

fully simulated digital environments. These advances in VR technology can, for example, be used to simulate AR content [110]. Recent research [56, 146] shows promises in this VR approach.

It has been more than five years since Gabbard et al.’s landmark research paper on driver challenges and opportunities for AR automotive applications [36]. In 2014, technologies for AR in vehicles were not yet technically mature for practical use in vehicles. Challenges that needed to be solved include computation and perception issues, such as accurate capturing and interpretation of road geometry, precise vehicle positioning, compensation for vibrations, driver monitoring and implementation of sophisticated algorithms to create precise augmentation content in the driver’s field of view, and others. Additionally, physical, cognitive, and experiential issues (e.g., focus change in AR systems causing eye strain; dissonance in equilibrium sense causing simulator sickness; change blindness; cognitive overload, driver and passenger experience) needed to be tackled as well [36]. We studied all AutoUI publications since 2014 and found that an increasing number of them utilize mixed reality technology for their user studies. Mixed reality technology in vehicles has been researched for several years now and has shown to have the ability to increase trust and safety in automated driving as well as foster comfort of driving [7, 13, 49, 67, 68, 130, 153].

Table 1 gives an overview of selected augmented and virtual reality (AR, VR) applications. VR is commonly used for prototyping, simulating pedestrian interactions and tele-operated driving in highly specialized scenarios, which could be considered unsafe in the real world. Additionally, passenger experiences are often modeled in VR. On the other hand, AR is researched for infotainment and gamification concepts in automated driving, as well as displaying navigation hints and hazard indications. More recently, mobile work and well-being are researched topics in the AR domain.

The 2019 International Conference on Automotive User Interfaces and Interactive Vehicular Application [1] was accompanied by a workshop on mixed reality in intelligent vehicles [108] with the purpose of finding a holistic approach of MR use in driving. In this paper, we present the results of the mixed reality workshop for intelligent vehicles with focus on the practical usage of mixed reality as well as future developments of this technology in intelligent vehicles, categorized into opportunities and challenges.

Figure 2 shows our brainstorming and discussion results. The main focus of the workshop was to identify challenges, subcategorized into *accessibility* (Figure 2b), *usability testing* (Figure 2c)

and *evaluation criteria* (Figure 2d) as well as *goals and opportunities* (Figure 2a), which are presented in detail in the following.

2 GOALS AND OPPORTUNITIES

Utilizing mixed reality technology can benefit drivers, passengers and pedestrians. One benefit of MR is to personalize content presentations (e.g., using a head-mounted display or a large 3D AR windshield display). In this section, we describe goals and opportunities that MR technology delivers in combination with automated vehicles.

2.1 Perception

An opportunity of MR technology used in intelligent vehicles is to overlap information on the windshield, providing the benefit for the driver of avoiding to look down to dashboard displays and keep focussed on the road [134]. However, it is necessary to accommodate the eyes between the outside environment and inside AR display. In this regard, also the distance to project information onto must be researched further [82, 84]. In partial automation, AR head-up or windshield displays provide assistance to human drivers by highlighting navigation and hazards [45, 143]. For example, adjusting the translucency of parts of the windshield display led to decreasing braking times [143], and highlighting the location of hazard on the WSD led to fewer shifts of visual focus and attention away from the road [45]. Additionally, it is possible to personalize the user interface using virtual/digital 3D objects rather than hardware buttons and knobs. Initial research conducted by Haeussel et al. [55] and Riegler et al. [113] shows that drivers of automated vehicles prefer personalized user interfaces tailored to their contextual situation (e.g., work-related or entertainment-related activities).

2.2 Multimodal Interaction and Arbitration

Employment of MR technology in vehicles additionally opens the possibility of new interaction modalities, i.e. how drivers and passenger can control the car and perform non-driving related activities. Such interaction types include gestures [116], speech [103], gaze [95, 107], haptic controls [47] etc. For example, Lagoo et al. [72] show that using gestures to interact with a HUD as opposed to a head-down display (HDD) mitigates driver’s distraction, ultimately leading to a reduction in collision occurrences. Moreover, there is an opportunity to wear HMDs, such as the Microsoft HoloLens, as opposed to built-in HUD displays [8]. Specifically, Benz et al.

[8] found that projection-based displays induced less simulator sickness in a real vehicle driving simulator. However, with future technological advancements of HMDs, especially when considering reduced weight, head-tracking and stereoscopic vision, HMDs might become a viable transport of information compared to HUDs.

2.3 Establish Guidelines and Recommendations

In traditional UI and HCI design, Shneiderman's Golden Rules [131] and Norman's Principles of Interaction Design [100] are well established. Consequently, when it comes to designing MR applications, guidelines and recommendations play an important role that may differ from the existing, traditional ones (e.g., designing for desktop and mobile devices). From tackling motion sickness to enhancing user experience and vehicle safety, such standardized "rules" must be developed with high priority. Interaction design guidelines could, for example, include the level of safety of MR interactions, the use of central vs. peripheral information systems, and a design space for MR applications [149]. The provision of UI AR widgets and tool kits can further help to create a uniform design (e.g., fonts, symbols, strokes) across different OEMs.

2.4 Support Non-Driving Related Task (NDRT) Experiences

For highly automated vehicles, non-driving related tasks (NDRTs) will replace the primary driving task. The transition from information displays nowadays to entertainment and experience displays in the future opens new possibilities for automotive UI designers. For example, using VR technology, the outside surroundings can be replaced with dynamically created digital content that adapts to the vehicle's driving behavior (direction, speed etc.) [87]. It is foreseeable that games and other entertainment experiences will be established in future vehicles. Gamification approaches can be used to establish trust in automated driving [50, 138, 148]. For example, Steinberger et al. [138] propose a combination of video game design theory and road safety psychology to engage drivers to drive safer while maintaining fun by displaying 3D AR objects on the windshield. Additionally, passengers in the back of the vehicle can be made aware of car activities using MR enabled side windows or displays. Co-operative AR user interfaces enable passengers in the vehicle and external vehicles to communicate with each other [79, 83]. Furthermore, pedestrian-vehicle communication can be achieved with MR technology using MR enabled windows and displays. The vehicle could augment its intention to pedestrians. For example, Loecken et al. [83] investigated how automated vehicles (AVs) should interact with pedestrians and found that displaying information on the street and using unambiguous signals (e.g., green lights) has a positive impact on pedestrian crossing safety and user experience. Additionally, MR can act as enabler for mobile office tasks as well [120]. Schmartmueller et al. [120] compared auditory and head-up displays for work-related tasks and found that the latter enable sequential multi-tasking as well as reduce workload and improve productivity.

3 CHALLENGES

The goal of our research roadmap is to address the challenges in HCI research with MR in automated driving. Therefore, the main focus of our workshop was to identify challenges regarding user study design and methods for evaluating MR applications and scenarios. In the following, we present in detail the challenges of applying MR technology in automated vehicles according to our workshop participants.

3.1 Challenge 1: Accessibility

One of the biggest challenges for provisioning mixed reality in automated vehicles is to make it accessible to their users. Drivers must trust MR and see the benefit of using it which requires context awareness and the ability to explain the driving situation, and if needed, provide take-over assistance.

3.1.1 Ensure Situation Awareness (SA). Mixed reality must be implemented in a way that provides drivers with a sufficient level of situation awareness [28]. To this end, the driver's status must be monitored: "Is the driver ready?" "Has the driver checked all objects nearby?" The intelligent vehicle must take appropriate actions should the driver "fall out of the loop" [115]. Riener et al. [115] state the need to identify how automotive HCI researchers can best monitor and measure the driver states, how they can provide appropriate feedback depending on the states and how they can estimate the driver's response to that feedback. Riener et al. [115] foresee a relaxation of visual display requirements in higher levels of vehicle automation (SAE level 3 and 4), as opposed to eyes-off-the-road or eyes-off-focus requirements for manual driving.

3.1.2 Interface Display. On the one hand, MR technology can be used to highlight danger objects from the outside environment on the interface display, such as a head-up or windshield display [45, 143]. Haeuslschmid et al. [45] propose to utilize AR windshield displays to visualize warnings at the position of the hazard and found that drivers employ a more calm gaze behavior (i.e., fewer shifts of visual focus and attention away from the road to detect an equivalent amount of hazards) when compared to the no display condition. On the other hand, a certain level of diminished reality can be achieved by using MR to adapt the interfaces to the current state of the driver and environment. Sawabe et al. [117, 118] aim to reduce motion sickness in automated driving by usingvection illusion to induce the user to make a preliminary movement against the real acceleration. Therefore, the intelligent vehicle must be aware of the driving context. Furthermore, using head-mounted displays within vehicles provides an opportunity to have the information display show private information for each occupant without violating their privacy.

3.1.3 Explainable UI. Regarding information transfer from the vehicle to the driver/ passengers, MR can be used to convey the intent of the vehicle: "What does the vehicle want to do?" This way, system knowledge and actions are made clear to the occupants of the vehicle. Explainable UI can be utilized for automated driving systems and shows whether the artificial intelligence (AI) is able to take over the driving control and the reasons for doing so [64]. MR projects what the car is "thinking" and gives an opportunity to the driver/passenger to decide the next action [71, 123]. Other

than for driving information, MR can be employed for work-related or entertainment purposes, such as immersive communication or video games [104]. Additionally, MR has the potential to provide pedestrians with information, such as route guidance [7].

3.1.4 Trust and Acceptance. MR can further be applied to leverage and increase trust in automated driving. We have to consider trust while keeping the driver in the loop. "When should the information flow be reduced?" Therefore, we must further investigate as to how trust can be managed with MR. For example, trust can be improved via display of behaviors of the vehicle, i.e., transparency of its actions [49, 152]. Wintersberger et al. [152] stress that as more and more fully automated vehicles emerge into our traffic systems, it will be very important to implement means for fostering user acceptance and trust. Wintersberger et al. [152] utilized AR aids in fully automated driving aiming to foster trust by increasing system transparency and communication of upcoming maneuvers and found that augmenting traffic objects relevant for a driving scenario can increase user trust and that users felt anxious without AR indicators. Similarly, Haeuslschmid et al. [49] use AR to visualize the car's interpretation of the current situation and its corresponding actions by means of a chauffeur avatar and a miniature representation of the road environment and found that such visualizations can increase trust in automated driving.

3.1.5 Risk Awareness. Especially for SAE level 3 and 4, MR technology can be applied to make the driver risk aware by pointing out the vehicle's level of confidence [126]. Schroeter et al. [126] present an approach to increase situational awareness by the means of AR to carefully design applications focused on amplification and voluntary attention. Their example application, Pokémon DRIVE, utilizes game design elements such as challenge, rewards, and narratives to direct driver attention to road hazards. The driver then needs to perform an interaction to acknowledge the AR hazard (e.g., swipe action on steering wheel). However, modeling risk awareness in terms of user interface design is challenging: "What objects from the outside environment should be visualized/highlighted by MR? All vehicles and pedestrians, or just moving vehicles and pedestrians intending to cross the street?"

3.1.6 Motion Sickness. One of the most important challenges when implementing MR in automated vehicles is to avoid motion sickness [30]. As Diels et al. [30] state, envisioned scenarios for self-driving cars will lead to an increased risk of motion sickness which will negatively affect safety and user acceptance. They further remark that self-driving cars cannot simply be thought of as living rooms on wheels. Since the primary driving task is no longer with the driver, it is essential to combat sickness occurring during NDRTs. Additionally, when using VR in automated vehicles, the overlay of virtual content must comply with the movement of the vehicle. Low latency between the real movement of the vehicle and the digital representation must therefore be ensured [42]. For automated vehicles, the use of MR provides the benefit to reduce motion sickness by showing navigation cues to the occupants of the vehicle, thereby allowing them to prepare for sharp turns or overtaking maneuvers. For example, McGill et al. [88] investigated the use of VR HMDs in automated driving scenarios to combat motion sickness. They found that there is no one best presentation with

respect to balancing sickness and immersion, however, according to user preferences, different solutions are required for differently susceptible users to provide usable in-vehicle VR.

3.1.7 Inattentional Blindness. The use of MR technology in the vehicle should not cause inattentional blindness, i.e. when the driver fails to perceive visual information that is relevant, detectable and within the useful field of view [62]. Especially in SAE level 3 driving scenarios, driver attention is critical and difficult to maintain. Therefore, designers of MR applications should strive for reducing unnecessary distractions for drivers. The ability for the driver to personalize their display(s) could influence inattentional blindness [55, 109, 114]. Haeuslschmid et al. [55] let drivers put together their own personalized layout (e.g., music player) in manual driving and found that personalized layouts resulted in less safe driving conditions (worse response rates and times). In SAE level 3 driving, drivers seem to be aware of the limitations of vehicle automation when it comes to potential take-overs and this circumstance is reflected in personalized layouts. Work-related information is placed in more peripheral areas of the windshield display in SAE level 3 driving, while it is placed more centrally in SAE level 5 vehicle automation [114].

3.1.8 Training. In addition, MR can facilitate driving training outside of real-world environments. For higher levels of vehicle automation (SAE level 3 and up), even a new driving license may be needed. Utilizing MR driving simulators has the benefit to reduce accident risks for beginners and simulate traffic scenarios that would be hard to find or reproduce in the real world. VR driving simulators have the ability to visualize various settings, from mountainous terrain to densely populated cities [9, 77, 112]. Several open source VR driving simulators with focus on HCI research were recently released to facilitate these research efforts [41, 110].

3.1.9 Take-Over. For take-over maneuvers, MR can also be applied [80, 111]. Lindemann et al. [80] examine the effects of using an AR interface with world-relative visualizations to assist drivers before a take-over and found that the AR interface, compared to a traditional head-down display, resulted in higher lateral performance and reduced workload in situations where steering is required directly after a take-over as well as better comfort of use and helpfulness. In case of partial vehicle automation, MR can also be employed to visualize take-over requests. In this regard, all recognized and relevant objects of the outside environment (e.g., cars, road, pedestrians) can be highlighted for the benefit of the driver to focus on the take-over. Schall et al. [119] visualize AR cues to increase driving safety among elderly drivers who are at higher crash risk because of cognitive impairments. Using these AR cues led to improved detection of hazardous target objects of low visibility while not interfering with detection of nonhazardous secondary objects. Further exploration of take-overs using MR, especially while drivers perform NDRTs, would therefore be advisable.

3.1.10 Communication with the Outside World. Moreover, MR applications can convey information to people outside and interaction with pedestrians [24, 48]. Examples are playful interactions (while pedestrians are waiting at a red light, [122]) or helpful information (e.g., helping tourists navigate to their destination; displaying a warning that danger is ahead, [29, 135]). MR technology enables

both private and shared spaces and thereby facilitates social connections. In addition to vehicle to pedestrian communication, pedestrians can communicate with each other via the MR technology of an intelligent vehicle. For example, the large windows/displays of vehicles enable pedestrians to share and visualize more information that by using smartphones. Vehicle to environment communication enables further use cases, such as notifying other road users (cars, cyclists etc.) of critical events (e.g., accidents, traffic jam).

3.1.11 Work, Entertainment and Well-Being. MR for work and entertainment purposes can be advantageous for highly automated driving (SAE level 3 and higher) [114, 151]. The outside environment could be completely overlaid with a virtual scene in real time. For example, instead of driving through a city, the passengers of the vehicle would see a visualization of a beautiful beach. For shared vehicles, MR technology enables information displays tailored to the personalization needs of each "user" of the vehicle. Instead of having a "one size fits all" solution, MR provides customizable displays according to the needs of the driver/passengers [55]. Haeuslschmid et al. [55] investigated driver needs for personalizable content on a windshield display in order to shift the urge of drivers to access content on smartphones while driving. Their findings show that drivers do prefer personalization, although such layouts may not be safe, which must be investigated further. Additionally, different visualization techniques using AR windshield displays might enable drivers to work more productively in a vehicular environment, when the digital content blends into the outside environment [111]. In an exploratory study by Riegler et al. [114], compared text comprehension and take-over performance on screen-fixed HUDs vs. world-relative displays and found that world-relative visualizations achieved better task performance and faster take-over times.

3.1.12 Hardware and Technological Limitations. Another challenge is the resolution of AR head-up displays which are currently small and offer limited space for information presentation. Microsoft recently released the HoloLens 2 HMD which has a vastly improved performance and field of view than its predecessor, making it more attractive for AR explorations in automated vehicle HCI research [147]. Moreover, the entire vehicle specification must be regarded, such as available (graphics) processors, bandwidth and other devices. A holistic approach towards hardware related solutions is therefore needed.

3.1.13 Laws and Regulations. Further challenges about the employment of MR technology include laws and regulations [132, 133] which limit the potential use cases for this technology, as well as the connection between researchers, hard- and software suppliers and OEMs. Smith [132, 133] presents steps that governments can take to encourage the development, deployment and use of automated road vehicles, as future mobility concepts emerge and research, development, demonstration, and deployment of these automated driving technologies accelerate.

3.2 Challenge 2: Usability Testing

Usability testing for MR scenarios can differ from traditional user studies, as many users are not yet familiar with MR devices, such as head-mounted displays or CAVEs and how to interact with them. In this section, we present methods and measures our workshop

participants came up with to adopt usability testing for mixed reality applications.

3.2.1 Methods. Methods for testing MR scenarios which have been suggested or applied so far include:

- Rapid Prototyping [5, 61, 75, 112, 136]
- Wizard of Oz [74, 94, 96]
- Testbeds [19, 58, 85, 86, 105, 106]
- Benchmark Scenarios [89]

MR concepts and use cases should be designed for each level of driving automation. The question arises if the same scenarios or different ones should be used when interacting with MR content. For example, do we have to define new interaction modalities with MR as opposed to traditional dashboard displays, for example. Additionally, user testing methods should be usable across different scenarios and domains. VR setups further simplify and speed up the prototyping process. Using VR technology, such content (VR applications) can easily be made universally available across research institutions and makes research more reproducible.

3.2.2 Data Sets. Data sets of (published) user studies should be made available as open source in order to support their reproducibility [21]. Going further, research institutions working together across studies can provide and access these data sets to more rapidly test methods and develop new ones. Additionally, open source provision of simulations and scenarios would be beneficial (e.g., [41, 112]). The consensus was that a coordinated approach can greatly speed up the HCI research process on MR scenarios (e.g., making MR passenger experiences openly available - which could simply be a Unity scene).

3.2.3 Simulations. Commonly, user studies on automated driving are either conducted using high fidelity driving simulators with motion platforms or even prototypical automated vehicles on test grounds, or low quality setups using 2D monitors. The question arises if currently available MR technology and fidelity can replace on-road evaluations, and what modalities are needed for good MR testing environments [9, 27, 77, 112, 124]. A number of (open-source) driving simulation testbeds has been developed with different goals, such as a 360° video-based driving simulator by Gerber and Schroeter [41, 124] or one with focus on windshield display and multimodal interaction research by Riegler et al. [110, 112]. Comparatively, access to MR equipment is cost-efficient and can therefore more easily be facilitated for HCI researchers than expensive driving simulators. Further, simulations using MR technology must be evaluated if they are also valid in real-world situations. For example, how does VR/simulated AR translate to "real" AR (e.g., depth, refocusing, parallax effect etc.). Goedicke et al. [42] developed a virtual reality simulator that can be used in real vehicles, thereby achieving higher levels of immersion. The main challenge for simulations is to find the level fidelity necessary for MR content to be considered valid.

3.2.4 Measures. User testing measures should be a mixed-modality design, applying method triangulation [102], such that multiple measures are considered at a time (e.g., self-developed/standardized questionnaires, activity logging and semi-structured interviews). Pettersson et al. [102] investigated 280 papers and found that method

triangulation was applied in more than two thirds of the user studies and concluded that, for stronger results in UX studies, collected data should be integrated and structured better. The consensus was that the following measures should be included in MR user tests:

- Motion or simulator sickness, e.g., using the Simulator Sickness Questionnaire (SSQ) [30, 63]
- Physiological measures, e.g., heart-rate variability and galvanic skin response [18]
- Workload, e.g., self-rating scales using the NASA Task Load Index (TLX) [18]
- Level of immersion or presence, e.g., using the Igroup Presence Questionnaire (IPQ) [15, 127]
- Oculometric measures, such as glance or gaze behavior [17, 95]

3.2.5 User Experience and Preference. User testing for MR applications should include user experience (UX) data and user preferences, however, preference is not necessarily equal to better or safer design. Therefore, safety aspects should also be evaluated in user studies. For example, depending on the level of vehicle automation [26], take-over request (TOR) performance could be evaluated [89]. Concrete TOR performance measures would be the reaction time, maximum longitudinal and lateral accelerations, and time to collision [43, 89]. Furthermore, MR technology should not be confined to in-vehicle driver/driving experiences, but also extended to passengers (e.g., [46]) and pedestrians (e.g., [44, 46, 83]). For example, Häkkinen et al. [46] propose a passenger concept for AR social car windows that focus on the surroundings, entertainment and social aspects of the journey. As for pedestrian user experiences, for example, Gruenefeld et al. [44] investigate different interaction modalities between vulnerable road users and automated vehicles, such as vehicles displaying their intentions or reacting to pedestrians' gestures. To this end, they created a VR testbed to evaluate such experiences.

3.3 Challenge 3: Evaluation Criteria

Evaluation criteria for MR scenarios differ from traditional user studies, as extra care and caution is needed for using and interacting with MR devices. Inexperienced MR users must be introduced to this technology, and simulator sickness could occur if the technology is not implemented correctly. In this section, we present evaluation criteria for mixed reality centered user studies.

3.3.1 Safety and Human State. While MR offers the possibility to increase safety, we must consider challenges like information density, cognitive overload and visual distractions. Keeping the driver in the loop [128] must therefore be a high priority for MR application in vehicles. Seppelt et al. [128] highlight the importance of continuous feedback on the state and behavior of automation as well as the feedback design needed to direct attention to informing rather than alerting drivers. Further, task performance as well as situation awareness measures should be defined for MR activities; i.e. driver response to hazards and glance behavior [28, 35]. De Winter et al. [28] investigated workload and SA during highly automated driving and found that drivers are likely to pick up tasks that are unrelated to driving. High vehicle automation can lead to improved SA compared to manual driving if drivers are encouraged

to detect objects in the environment. In this context, MR can be exploited to increase the driver's situation awareness and take-over performance (e.g., [60, 95, 101]). For example, Langlois et al. [73] compare AR and classical HUDs displaying navigational information and found that AR both improves driving comfort and helps better anticipate lane change maneuvers when performing take-overs. Riegler et al. [111] found that world-relative visualization of (work- or entertainment related) text content can improve task and take-over performance compared to traditional screen-fixed HUDs.

3.3.2 User-centered Criteria. Regarding user-centered criteria, trust and acceptance of MR interactions, usability and user experience were regarded as important. Further criteria include previous experience with this technology and in this regard the "wow" effect of MR. Additionally, MR enables adaptive user interfaces based on different user profiles and preferences. User tech-savviness plays an important role with MR research, as study participants should not be immediately confronted with this technology, but first introduced and given the chance to experience it before conducting user studies [22]. "Warm-up" strategies to familiarize users with the MR environment are therefore recommended [22]. Moreover, even when conducting user studies in real vehicles, utilizing VR environments prior to the real environment has shown to yield a better performance [22].

3.3.3 Simulator and Motion Sickness. When conducting user studies on MR in intelligent vehicles, simulator and motion sickness must be considered. Motion sickness may occur when stationary users experience self-motion or when lags between head movements and presentation of the visual display take place [51]. Prior to conducting a user study, a pre-screening should be conducted to test if a study participant has existing issues, such as disorientation or nausea [147]. To assess simulator sickness among study participants, the simulator sickness questionnaire (SSQ) can be used [63]. To this end, the duration and immersion of user studies must carefully be planned, as these factors may lead to nausea, fatigue, headache, dizziness, etc. [97]. Murata [97] found that longer immersion in a VR environment induced postural instability and symptoms of motion sickness. Consequently, adaptive UIs could be used to limit motion sickness [33]. For example, Draper et al. [33] examined simulator sickness in virtual environments and found that the vestibular system is designed to detect and react to the position and motion of the head in space. When creating VR simulations, it is therefore necessary to coordinate motor function, ensure correct eye movements and maintain posture in the virtual environment in order to reduce health and safety issues [33]. Should motion or simulator sickness occur, the cause of these effects (e.g., hardware issues or the actual concepts) must be identified. Therefore, the level of immersion and presence should be as appropriately high: Closeness of the scenarios to real life situations makes them representable and universally applicable [16]. As Bowman and McMahan [16] state, full immersion is not always necessary; the goal of immersive virtual environments is to let the user experience a computer-generated world as if it were real, thereby producing a sense of presence.

3.3.4 Hardware. Additionally, not just for evaluating motion or simulator sickness, but also for the level of realism, the physical car

state (e.g., acceleration, sounds), immersion and presence should be as consistent and generalizable to reality as possible [16]. Bowman and McMahan [16] propose a multidimensional approach for immersion and presence in VR in which VR designers should strive for a balance of spatial understanding, information clutter and peripheral awareness. This notion should especially be emphasized for VR studies on AR or MR usage, as hardware devices and setups (e.g., head-mounted displays, CAVEs) have limitations (e.g., resolution, field of view, refresh rate, display technology). Multi-sensory tangibles can be used support direct interaction with the real world by enabling the use of real, physical objects and tools [93]. In VR driving simulators, for example, this could be achieved using a hardware steering wheel and pedals in combination with a virtual representation of the user's hands [110]. Furthermore, MR concepts should be coherent and holistic rather than multiple single concepts put together, even though such an approach makes it harder to isolate effects or evaluate the interference of factors (e.g., motion sickness, user experience, performance).

4 RESEARCH ROADMAP

Researchers working in the automotive UI domain may further explore mixed reality technology in the context of automated driving. However, there is still a number of open issues to clarify prior to a broad application of AR technology in vehicles, e.g., questions addressing the use of head-mounted displays or the utilization of windshield displays. Also, topics like context awareness, information relevancy as well as view management are research topics of interest. In addition, further interaction modalities in combination with AR content should be researched for the creation of natural vehicle user interfaces (e.g., finger/ hand gestures, speech commands, gaze input). Vehicle-to-vehicle or pedestrian-to-vehicle interaction gained lot of interest recently and should be further investigated in the light of MR (e.g., communication strategies in the exterior). The technical progress also extends the use cases of mixed reality applications with implications on new human factors research fields, e.g., passenger or entertaining and work experiences.

Gartner's hype cycle [37] is a graphical representation of a common pattern that occurs with emerging technologies, showing the current maturity status of these technologies. The hype cycle can estimate the growth or depletion of a given technology. Back in 2014, both augmented and virtual reality were in the phase of "Trough of Disillusionment", meaning that these technologies started to be rigorously tested with experiments and implementations [38]. In 2019, both AR and VR had disappeared from the Gartner hype cycle for emerging technologies [39], indicating, that for Gartner, mixed reality was not an emerging and experimental technology anymore, but it had already matured, i.e., become usable and useful. However, autonomous driving at SAE level 4 has reached the "Trough of Disillusionment". For AutomotiveUI HCI researchers, this implies that MR will enter (partially) vehicles in the foreseeable future, and, with its goals and opportunities in mind, the mentioned challenges must be addressed swiftly. For example, Daimler has recently announced and showcased an AR HUD which will display world-relative navigational cues as well as vehicle-related information to the driver as part of their MBUX (Mercedes-Benz User Experience) suite [2].

Our workshop and subsequent investigations have several implications for future mixed reality applications for automated driving. Based on the presented opportunities and challenges, we define the following overarching action points for HCI researchers and AR/VR designers in the automotive domain:

- **View management** for MR applications (SAE L3-L5). MR technology paired with automated driving open the prospect to transition away from traditional hardware-based knobs and buttons to application-specific UI elements. AR head-up and windshield displays may be an enabler for minimalistic vehicle interiors and app-centered user experiences.
- **Interaction modalities** for MR-based NDRTs (SAE L3-L5), i.e., a holistic rather than a standalone approach must be explored for interacting with MR applications. A holistic approach would include gaze, gestural, haptic, speech/auditory, olfactory and more interaction and feedback types. Using such an integrated approach in combination with the proper view management has the potential to improve the driver's situation awareness and reduce motion sickness.
- **Ensuring situation awareness** while performing NDRTs (SAE L3 and L4). It is necessary to consistently monitor the driver state (e.g., using physiological measures) to allow for fast and successful take-overs. The driver needs to be aware of potential hazards. In this regard, context awareness must be investigated further to ensure both safe and pleasant transitions from automated to manual driving.
- **Passenger experiences.** The shift from manual to automated driving transforms active drivers to passive passengers. The use of AR glasses and VR HMDs should be explored further for combating motion sickness and creating practical in-car work- and entertainment related experiences. Gamification concepts should be researched further to explore their potential in increasing the trust, acceptance and eventually popularity in automated vehicles.
- **Pedestrian interactions.** External vehicle interfaces must be researched further to allow safe interaction with vulnerable road users, such as pedestrians and cyclists.
- **Simulation and testing.** Due to the unlimited number of potential MR applications for in-vehicle and external usage, safety and usability testing criteria as well as simulations must be defined. Data sets should be made available to HCI researchers to ensure continuous development and improvement of these criteria.

5 CONCLUSION

Mixed reality applications for in-vehicle driver and passenger experiences as well as external pedestrian experiences are near a turning point. Usable MR display technology for the entertainment industry is already available, and recent prototypes of windshield-based optical see-through displays showcased by automotive manufacturers and suppliers in combination with higher levels of vehicle automation may facilitate commercially available systems within the next few years. The most successful automotive MR applications will have to consider a wide range of perceptual and cognitive issues within the (semi-automated) driving context. We present some of

these challenges based on our expert workshop, and outline a research agenda. However, much more research is needed to ensure safe and practical MR applications for future mobility.

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